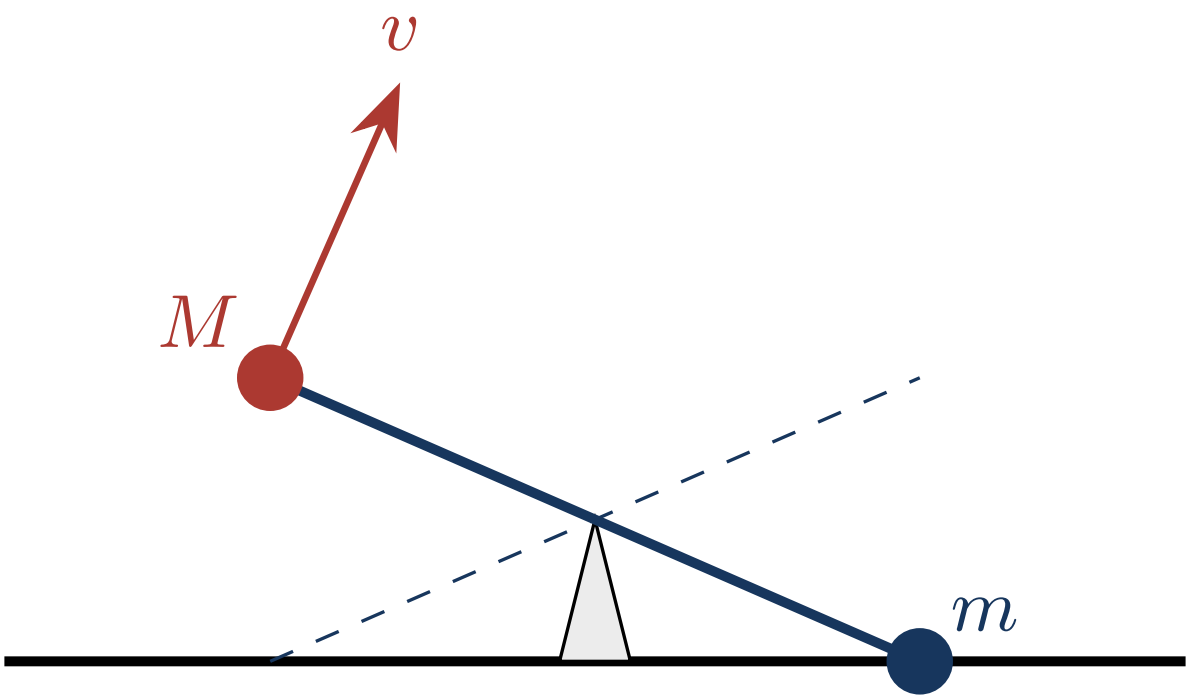$$\Delta U = 2Hg(M - m)$$